

CS 2601 Linear and Convex Optimization

15. Lagrange duality in general problems

Bo Jiang

John Hopcroft Center for Computer Science
Shanghai Jiao Tong University

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Outline

- Dual function and dual problem
- Weak and strong duality
- Slater's condition
- KKT conditions revisited

Lagrange dual function

Consider the general optimization problem (not necessarily convex),

$$\begin{aligned} \min_{\mathbf{x}} \quad & f(\mathbf{x}) \\ \text{s.t.} \quad & h_i(\mathbf{x}) = 0, \quad i = 1, 2, \dots, k \\ & g_j(\mathbf{x}) \leq 0, \quad j = 1, 2, \dots, m \end{aligned} \tag{P}$$

The Lagrangian is

$$\mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\mu}) = f(\mathbf{x}) + \sum_{i=1}^k \lambda_i h_i(\mathbf{x}) + \sum_{j=1}^m \mu_j g_j(\mathbf{x})$$

The (Lagrange) dual function is

$$\phi(\boldsymbol{\lambda}, \boldsymbol{\mu}) = \inf_{\mathbf{x} \in D} \mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\mu}) = \inf_{\mathbf{x} \in D} \left(f(\mathbf{x}) + \sum_{i=1}^k \lambda_i h_i(\mathbf{x}) + \sum_{j=1}^m \mu_j g_j(\mathbf{x}) \right)$$

where $D = \text{dom} f \cap (\bigcap_{i=1}^k \text{dom} h_i) \cap (\bigcap_{j=1}^m \text{dom} g_j)$ is the domain of the problem. We will downplay the role of D and focus on the case $D = \mathbb{R}^n$.

Example

Given $\mathbf{A} \in \mathbb{R}^{k \times n}$,

$$\begin{aligned} \min_{\mathbf{x}} \quad & f(\mathbf{x}) = \|\mathbf{x}\|^2 = \mathbf{x}^T \mathbf{x} \\ \text{s.t.} \quad & \mathbf{A}\mathbf{x} = \mathbf{b} \\ & \mathbf{x} \geq \mathbf{0} \end{aligned}$$

The Lagrangian is

$$\mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\mu}) = \mathbf{x}^T \mathbf{x} + \boldsymbol{\lambda}^T (\mathbf{A}\mathbf{x} - \mathbf{b}) - \boldsymbol{\mu}^T \mathbf{x}$$

Since $\mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\mu})$ is convex in $\mathbf{x}$, its minimum satisfies

$$\nabla_{\mathbf{x}} \mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\mu}) = 2\mathbf{x} + \mathbf{A}^T \boldsymbol{\lambda} - \boldsymbol{\mu} = \mathbf{0} \implies \mathbf{x} = \frac{1}{2}(\boldsymbol{\mu} - \mathbf{A}^T \boldsymbol{\lambda})$$

The dual function is

$$\phi(\boldsymbol{\lambda}, \boldsymbol{\mu}) = \mathcal{L}\left(\frac{1}{2}(\boldsymbol{\mu} - \mathbf{A}^T \boldsymbol{\lambda}), \boldsymbol{\lambda}, \boldsymbol{\mu}\right) = -\frac{1}{4}\|\boldsymbol{\mu} - \mathbf{A}^T \boldsymbol{\lambda}\|^2 - \mathbf{b}^T \boldsymbol{\lambda}$$

Example

Given $A \in \mathbb{R}^{k \times n}$,

$$\begin{array}{ll} \min_x & f(\mathbf{x}) = \mathbf{x}^T \mathbf{x} \\ \text{s.t.} & A\mathbf{x} = \mathbf{b} \end{array}$$

The Lagrangian is

$$\mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}) = \mathbf{x}^T \mathbf{x} + \boldsymbol{\lambda}^T (A\mathbf{x} - \mathbf{b})$$

Since $\mathcal{L}(\mathbf{x}, \boldsymbol{\lambda})$ is convex in $\mathbf{x}$, its minimum satisfies

$$\nabla_{\mathbf{x}} \mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}) = 2\mathbf{x} + A^T \boldsymbol{\lambda} = \mathbf{0} \implies \mathbf{x} = -\frac{1}{2} A^T \boldsymbol{\lambda}$$

The dual function is

$$\phi(\boldsymbol{\lambda}) = \mathcal{L}\left(-\frac{1}{2} A^T \boldsymbol{\lambda}, \boldsymbol{\lambda}\right) = -\frac{1}{4} \boldsymbol{\lambda}^T A A^T \boldsymbol{\lambda} - \mathbf{b}^T \boldsymbol{\lambda} = -\frac{1}{4} \|A^T \boldsymbol{\lambda}\|^2 - \mathbf{b}^T \boldsymbol{\lambda}$$

Lower bound for optimal value

For any λ and any $\mu \geq \mathbf{0}$, the optimal value f^* of (P) is bounded by

$$f^* \geq \phi(\lambda, \mu)$$

Proof. Let $X = \{\mathbf{x} : h_i(\mathbf{x}) = 0, \forall i; g_j(\mathbf{x}) \leq 0, \forall j\}$ be the feasible set.

- If $X = \emptyset$, then $f^* = +\infty$, trivially true.
- If $X \neq \emptyset$, for $\mu \geq \mathbf{0}$ and $\mathbf{x} \in X$,

$$f(\mathbf{x}) \geq f(\mathbf{x}) + \sum_{i=1}^k \lambda_i \underbrace{h_i(\mathbf{x})}_{=0} + \sum_{j=1}^m \underbrace{\mu_j g_j(\mathbf{x})}_{\leq 0} = \mathcal{L}(\mathbf{x}, \lambda, \mu)$$

Minimizing over $\mathbf{x}$,

$$f^* = \inf_{\mathbf{x} \in X} f(\mathbf{x}) \geq \inf_{\mathbf{x} \in X} \mathcal{L}(\mathbf{x}, \lambda, \mu) \geq \inf_{\mathbf{x}} \mathcal{L}(\mathbf{x}, \lambda, \mu) = \phi(\lambda, \mu)$$

Concavity of dual function

The dual function is always concave, whether the primal problem (P) is convex or not.

Proof. Note $\mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\mu})$ is affine in $(\boldsymbol{\lambda}, \boldsymbol{\mu})$. Thus $\phi(\boldsymbol{\lambda}, \boldsymbol{\mu}) = \inf_{\mathbf{x}} \mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\mu})$ is the pointwise infimum of a family of affine functions indexed by $\mathbf{x}$, and hence concave. (Recall the pointwise supremum of convex functions is convex).

$$\begin{aligned}\phi(\boldsymbol{\lambda}, \boldsymbol{\mu}) &= \inf_{\mathbf{x} \in D} \left(f(\mathbf{x}) + \sum_{i=1}^k \lambda_i h_i(\mathbf{x}) + \sum_{j=1}^m \mu_j g_j(\mathbf{x}) \right) \\ &= - \underbrace{\sup_{\mathbf{x} \in D} \left(-f(\mathbf{x}) - \sum_{i=1}^k \lambda_i h_i(\mathbf{x}) - \sum_{j=1}^m \mu_j g_j(\mathbf{x}) \right)}_{\text{pointwise supremum of convex (affine) functions in } (\boldsymbol{\lambda}, \boldsymbol{\mu})}\end{aligned}$$

Example. $\phi(\boldsymbol{\lambda}, \boldsymbol{\mu}) = -\frac{1}{4} \|\boldsymbol{\mu} - \mathbf{A}^T \boldsymbol{\lambda}\|^2 - \mathbf{b}^T \boldsymbol{\lambda}$ is concave.

Lagrange dual problem

To find the best lower bound given by the dual function

$$f^* \geq \phi(\boldsymbol{\lambda}, \boldsymbol{\mu})$$

solve the (Lagrange) dual problem associated with the primal problem (P),

$$\begin{aligned} \max_{\boldsymbol{\lambda}, \boldsymbol{\mu}} \quad & \phi(\boldsymbol{\lambda}, \boldsymbol{\mu}) \\ \text{s.t.} \quad & \boldsymbol{\mu} \geq \mathbf{0} \end{aligned} \tag{D}$$

The dual problem (D) is **always** convex, whether or not (P) is convex.

$(\boldsymbol{\lambda}, \boldsymbol{\mu})$ is **dual feasible** if $\boldsymbol{\mu} \geq \mathbf{0}$ and $\phi(\boldsymbol{\lambda}, \boldsymbol{\mu}) > -\infty$.

Note. The domain of a convex function f is $\text{dom} f = \{\mathbf{x} : f(\mathbf{x}) < +\infty\}$, while the domain of a concave function f is $\text{dom} f = \{\mathbf{x} : f(\mathbf{x}) > -\infty\}$. Thus the condition $\phi(\boldsymbol{\lambda}, \boldsymbol{\mu}) > -\infty$ just means $(\boldsymbol{\lambda}, \boldsymbol{\mu}) \in \text{dom } \phi$.

Example

Recall the dual problem of the following LP

$$\begin{aligned} \min_{\mathbf{x}} \quad & f(\mathbf{x}) = \mathbf{c}^T \mathbf{x} \\ \text{s.t.} \quad & \mathbf{Ax} = \mathbf{b} \\ & \mathbf{Gx} \geq \mathbf{h} \end{aligned}$$

is

$$\begin{aligned} \max_{\boldsymbol{\lambda}, \boldsymbol{\mu}} \quad & \phi(\boldsymbol{\lambda}, \boldsymbol{\mu}) = \begin{cases} \boldsymbol{\lambda}^T \mathbf{b} + \boldsymbol{\mu}^T \mathbf{h}, & \text{if } \mathbf{A}^T \boldsymbol{\lambda} + \mathbf{G}^T \boldsymbol{\mu} = \mathbf{c} \\ -\infty, & \text{otherwise} \end{cases} \\ \text{s.t.} \quad & \boldsymbol{\mu} \geq \mathbf{0} \end{aligned}$$

$(\boldsymbol{\lambda}, \boldsymbol{\mu})$ is dual feasible if $\boldsymbol{\mu} \geq \mathbf{0}$ and $\mathbf{A}^T \boldsymbol{\lambda} + \mathbf{G}^T \boldsymbol{\mu} = \mathbf{c}$, which just means it is feasible for the dual LP,

$$\begin{aligned} \max_{\boldsymbol{\lambda}, \boldsymbol{\mu}} \quad & \psi(\boldsymbol{\lambda}, \boldsymbol{\mu}) = \boldsymbol{\lambda}^T \mathbf{b} + \boldsymbol{\mu}^T \mathbf{h} \\ \text{s.t.} \quad & \mathbf{A}^T \boldsymbol{\lambda} + \mathbf{G}^T \boldsymbol{\mu} = \mathbf{c} \\ & \boldsymbol{\mu} \geq \mathbf{0} \end{aligned}$$

Example

Recall the dual problem of the following problem

$$\begin{aligned} \min_{\mathbf{x}} \quad & f(\mathbf{x}) = \mathbf{x}^T \mathbf{x} \\ \text{s.t.} \quad & \mathbf{A}\mathbf{x} = \mathbf{b} \\ & \mathbf{x} \geq \mathbf{0} \end{aligned}$$

is

$$\begin{aligned} \max_{\boldsymbol{\lambda}, \boldsymbol{\mu}} \quad & \phi(\boldsymbol{\lambda}, \boldsymbol{\mu}) = -\frac{1}{4} \|\boldsymbol{\mu} - \mathbf{A}^T \boldsymbol{\lambda}\|^2 - \mathbf{b}^T \boldsymbol{\lambda} \\ \text{s.t.} \quad & \boldsymbol{\mu} \geq \mathbf{0} \end{aligned}$$

$(\boldsymbol{\lambda}, \boldsymbol{\mu})$ is dual feasible if $\boldsymbol{\mu} \geq \mathbf{0}$, as there is no implicit constraint in ϕ .

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Weak and strong duality

Denote by f^* and ϕ^* the primal and dual optimal values, i.e.

$$f^* = \inf_{\mathbf{x} \in X} f(\mathbf{x}), \quad \phi^* = \sup_{\boldsymbol{\lambda}, \boldsymbol{\mu}: \boldsymbol{\mu} \geq \mathbf{0}} \phi(\boldsymbol{\lambda}, \boldsymbol{\mu})$$

Weak duality: $f^* \geq \phi^*$

- **always** holds.

Proof. Recall $f^* \geq \phi(\boldsymbol{\lambda}, \boldsymbol{\mu})$ for any $\boldsymbol{\lambda}$ and any $\boldsymbol{\mu} \geq \mathbf{0}$. Weak duality follows by maximizing over $\boldsymbol{\lambda}$ and $\boldsymbol{\mu} \geq \mathbf{0}$.

- $f^* - \phi^*$ is called the (optimal) duality gap of the problem.

Strong duality: $f^* = \phi^*$

- does **not** hold in general.
- typically holds for convex problems under various conditions known as **constraint qualifications**, e.g. Slater's condition.
- may also hold for nonconvex problems.
- can solve the dual problem instead if it is easier than the primal.

Duality gap

Given primal feasible x and dual feasible (λ, μ) , the difference

$$f(x) - \phi(\lambda, \mu)$$

is called the **duality gap** associated with x and (λ, μ) .

Note

$$\phi(\lambda, \mu) \leq \phi^* \leq f^* \leq f(x)$$

If the duality gap is zero, i.e. $f(x) = \phi(\lambda, \mu)$, then all inequalities become equalities, so x is primal optimal, and (λ, μ) is dual optimal.

If the gap $f(x) - \phi(\lambda, \mu) \leq \epsilon$, then the dual solution (λ, μ) serves as a **proof** or **certificate** that x is ϵ -suboptimal,

$$f(x) - f^* \leq f(x) - \phi(\lambda, \mu) \leq \epsilon$$

When strong duality holds, this can serve as a stopping criterion in an iterative algorithm, i.e. stop when $f(x) - \phi(\lambda, \mu) \leq \epsilon$ for some (λ, μ) .

Example

$$\begin{array}{ll}\min_{x \in \mathbb{R}} & f(x) = x^2 \\ \text{s.t.} & x \leq a\end{array}$$

The dual function is

$$\phi(\mu) = \inf_x [x^2 + \mu(x - a)] = -\frac{\mu^2}{4} - a\mu$$

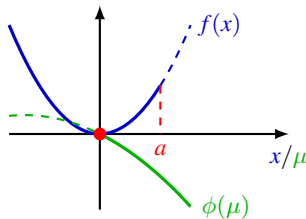
The dual problem is

$$\begin{array}{ll}\max_{\mu \in \mathbb{R}} & \phi(\mu) = -\frac{\mu^2}{4} - a\mu \\ \text{s.t.} & \mu \geq 0\end{array}$$

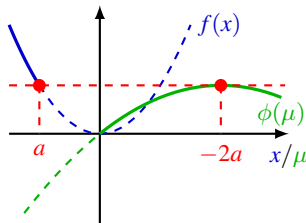
The primal and dual optimal values are

1. If $a \geq 0$, $f^* = f(0) = \phi^* = \phi(0) = 0$
2. If $a \leq 0$, $f^* = f(a) = \phi^* = \phi(-2a) = a^2$

Strong duality holds in both cases.



Case 1. $a \geq 0$



Case 2. $a \leq 0$

Example

Consider

$$\begin{array}{ll}\min_{x \in \mathbb{R}} & f(x) = x^3 \\ \text{s.t.} & x \geq 0\end{array}$$

The optimal value is $f^* = f(0) = 0$.

The dual function is

$$\phi(\mu) = \inf_x [x^3 - \mu x] = -\infty$$

so the dual optimal value is

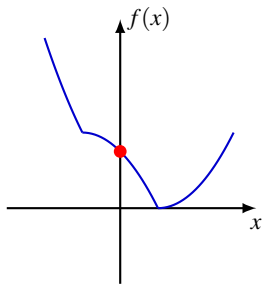
$$\phi^* = \sup_{\mu \geq 0} \phi(\mu) = -\infty$$

The duality gap is infinite. In particular, strong duality does not hold.

Example

Consider

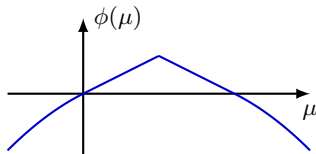
$$\begin{aligned} \min_{x \in \mathbb{R}} \quad & f(x) = \begin{cases} -x^2 - x + \frac{3}{4}, & |x| \leq \frac{1}{2} \\ x^2 - x + \frac{1}{4}, & |x| \geq \frac{1}{2} \end{cases} \\ \text{s.t.} \quad & x \leq 0 \end{aligned}$$



The primal optimal value is $f^* = f(0) = \frac{3}{4}$.

The dual function is

$$\phi(\mu) = \inf_x [f(x) + \mu x] = \begin{cases} \frac{1 - |\mu - 1|}{2}, & |\mu - 1| \leq 1 \\ \frac{1 - (\mu - 1)^2}{4}, & |\mu - 1| \geq 1 \end{cases}$$



The dual optimal value is $\phi^* = \phi(1) = \frac{1}{2}$.

The duality gap is $f^* - \phi^* = \frac{1}{4}$.

Example (cont'd)

To compute the dual function, note

$$\mathcal{L}(x, \mu) = f(x) + \mu x = \begin{cases} -x^2 + (\mu - 1)x + \frac{3}{4}, & |x| \leq \frac{1}{2} \\ x^2 + (\mu - 1)x + \frac{1}{4}, & |x| > \frac{1}{2} \end{cases}$$

Since $y = -x^2 + (\mu - 1)x + \frac{3}{4}$ is a parabola opening down,

$$\phi_1(\mu) = \inf_{|x| \leq \frac{1}{2}} \mathcal{L}(x, \mu) = \min \left\{ \mathcal{L}\left(\frac{1}{2}, \mu\right), \mathcal{L}\left(-\frac{1}{2}, \mu\right) \right\} = \frac{1 - |\mu - 1|}{2}$$

Since $y = x^2 + (\mu - 1)x + \frac{1}{4}$ is a parabola opening up,

$$\phi_2(\mu) = \inf_{|x| \geq \frac{1}{2}} \mathcal{L}(x, \mu) = \begin{cases} \mathcal{L}\left(\frac{1-\mu}{2}, \mu\right) = \frac{1-(\mu-1)^2}{4}, & |\mu - 1| \geq 1 \\ \min \left\{ \mathcal{L}\left(\frac{1}{2}, \mu\right), \mathcal{L}\left(-\frac{1}{2}, \mu\right) \right\} = \frac{1-|\mu-1|}{2}, & |\mu - 1| \leq 1 \end{cases}$$

Thus

$$\phi(\mu) = \min\{\phi_1(\mu), \phi_2(\mu)\} = \phi_2(\mu)$$

Example (cont'd)

By definition of dual function,

$$\phi(\mu) = \inf_x [f(x) + \mu x] \leq f(x) + \mu x$$

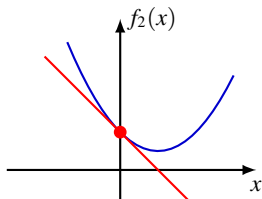
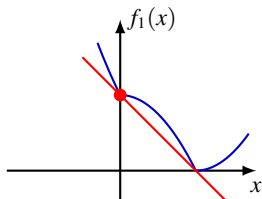
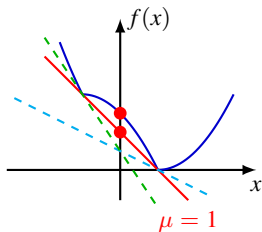
Rearranging,

$$\ell(x) \triangleq -\mu x + \phi(\mu) \leq f(x)$$

Note $\ell(x)$ is a line with slope $-\mu$ and intercept $\phi(\mu)$ that supports $\text{epi} f$.

The dual optimal value ϕ^* is the largest intercept of such lines. We can see pictorially there is a gap.

This also give us intuition about why strong duality may hold for nonconvex problem, and why it usually holds for convex problems.



Example

$$\min_{x \in \mathbb{R}} f(x) = \begin{cases} -\sqrt{x}, & x > 0 \\ 1 & x = 0 \\ +\infty, & x < 0 \end{cases}$$

$$\text{s.t. } x \leq 0$$

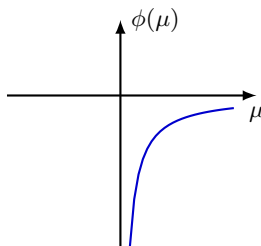
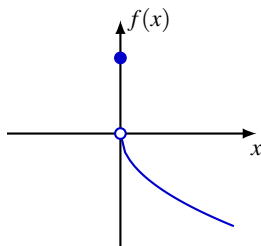
The primal optimal value is $f^* = f(0) = 1$.

The dual function is

$$\phi(\mu) = \inf_x [f(x) + \mu x] = \begin{cases} -\frac{1}{4\mu}, & \mu > 0 \\ -\infty, & \mu \leq 0 \end{cases}$$

The dual optimal value is $\phi^* = 0$, which is not attainable.

This is a convex problem with nonzero duality gap $f^* - \phi^* = 1$, a nontypical case.



Example

$$\min_{x \in \mathbb{R}} f(x) = \begin{cases} -\sqrt{x}, & x > 0 \\ 1 & x = 0 \\ +\infty, & x < 0 \end{cases}$$

$$\text{s.t. } x \leq a$$

where $a > 0$.

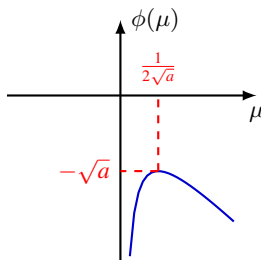
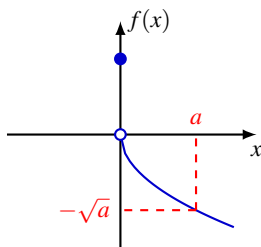
The primal optimal value is $f^* = f(a) = -\sqrt{a}$.

The dual function is

$$\phi(\mu) = \inf_x [f(x) + \mu(x-a)] = \begin{cases} -\frac{1}{4\mu} - a\mu, & \mu > 0 \\ -\infty, & \mu \leq 0 \end{cases}$$

The dual optimal value is $\phi^* = \phi(\frac{1}{2\sqrt{a}}) = -\sqrt{a}$

Strong duality holds in this case.



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Slater's condition for convex problems

Consider a convex problem,

$$\begin{array}{ll}\min_{\mathbf{x}} & f(\mathbf{x}) \\ \text{s.t.} & g_j(\mathbf{x}) \leq 0, \quad j = 1, 2, \dots, m \\ & \mathbf{h}(\mathbf{x}) = \mathbf{A}\mathbf{x} - \mathbf{b} = \mathbf{0}\end{array} \quad (\text{CP})$$

with domain $D = \text{dom} f \cap (\bigcap_{j=1}^m \text{dom} g_j)$.

Slater's condition. The above problem is **strictly feasible**, i.e.

$$\exists \mathbf{x} \in \text{int } D^1 \quad \text{s.t.} \quad g_j(\mathbf{x}) < 0 \text{ for } j = 1, 2, \dots, m, \quad \mathbf{A}\mathbf{x} = \mathbf{b}$$

Refined Slater's condition. If some g_j are affine, the requirement $g_j(\mathbf{x}) < 0$ can be relaxed to feasibility $g_j(\mathbf{x}) \leq 0$ for those g_j .

¹ $\text{int } D$ stands for the interior of D . $\mathbf{x} \in \text{int } D$ if there exists $\delta > 0$ s.t. $B(\mathbf{x}, \delta) \subset D$.
Again we focus on the case $D = \mathbb{R}^n$, so the requirement $\mathbf{x} \in \text{int } D$ is always satisfied. ²¹

Slater's Theorem

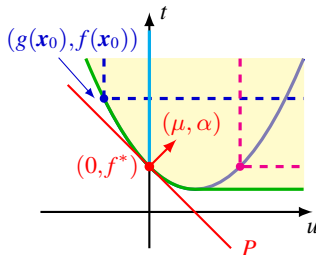
Slater's Theorem. Strong duality holds for (CP) under (refined) Slater's condition. Furthermore, if $\phi^* > -\infty$, it is attained by some (λ^*, μ^*) .

Proof idea. Consider the case with only one inequality constraint g . Let

$$C = \{(u, t) : \exists \mathbf{x} \in D \text{ s.t. } g(\mathbf{x}) \leq u, f(\mathbf{x}) \leq t\}$$

C is convex and $(0, f(\mathbf{x})) \in C$ for feasible $\mathbf{x}$. C has a supporting hyperplane P at $(0, f^*) \in \partial C$,

$$\mu u + \alpha t \geq \alpha f^*, \quad \forall (u, t) \in C$$



Letting $u, t \rightarrow +\infty$ shows $\mu \geq 0, \alpha \geq 0$. By Slater's condition, $\exists \mathbf{x}_0 \in D$ s.t. $g(\mathbf{x}_0) < 0$, so P is non-vertical, i.e. $\alpha \neq 0$. Since $(g(\mathbf{x}), f(\mathbf{x})) \in C$,

$$f^* \leq \mu^* g(\mathbf{x}) + f(\mathbf{x}) = \mathcal{L}(\mathbf{x}, \mu^*), \text{ where } \mu^* = \frac{\mu}{\alpha} \geq 0$$

Minimizing over $\mathbf{x} \in D$, $f^* \leq \phi(\mu^*) \leq \phi^*$. Weak duality then implies $f^* = \phi(\mu^*) = \phi^*$. The condition $\mathbf{x}_0 \in \text{int } D$ will be used to deal with h .

Proof

If $f^* = -\infty$, then $\phi^* = f^* = -\infty$ by weak duality. Assume $f^* > -\infty$. Since (CP) is strictly feasible by Slater's condition, $f^* < +\infty$. Also assume $A \in \mathbb{R}^{k \times n}$ has $\text{rank } A = k$, by removing redundant constraints. Now let

$$C = \{(\mathbf{u}, \mathbf{v}, t) : \exists \mathbf{x} \in D \text{ s.t. } \mathbf{g}(\mathbf{x}) \leq \mathbf{u}, \mathbf{h}(\mathbf{x}) = \mathbf{v}, f(\mathbf{x}) \leq t\}$$

1. C is convex. The proof is similar to that on slide 7 of §4 part 2.
2. $C \neq \emptyset$ and $(\mathbf{0}, \mathbf{0}, f^*) \in \partial C$.
 - 2.1 Note $(\mathbf{0}, \mathbf{0}, t) \in C$ iff $\exists \mathbf{x} \in D$ s.t. $\mathbf{g}(\mathbf{x}) \leq \mathbf{0}, \mathbf{h}(\mathbf{x}) = \mathbf{0}, f(\mathbf{x}) \leq t$, i.e. iff $\exists \mathbf{x} \in X$ s.t. $f(\mathbf{x}) \leq t$. In particular, $(\mathbf{0}, \mathbf{0}, f(\mathbf{x})) \in C$ for $\mathbf{x} \in X$.
 - 2.2 Since $f^* = \inf_{\mathbf{x} \in X} f(\mathbf{x})$, there exists a sequence $\{\mathbf{x}_i\} \subset X$ s.t. $f(\mathbf{x}_i) \rightarrow f^*$. Since $C \ni (\mathbf{0}, \mathbf{0}, f(\mathbf{x}_i)) \rightarrow (\mathbf{0}, \mathbf{0}, f^*)$, we have $(\mathbf{0}, \mathbf{0}, f^*) \in \overline{C}$.
 - 2.3 $(\mathbf{0}, \mathbf{0}, t) \notin C$ for any $t < f^*$. Thus $(\mathbf{0}, \mathbf{0}, f^*) \notin \text{int } C$ and $(\mathbf{0}, \mathbf{0}, f^*) \in \partial C$.
3. There exists a supporting hyperplane at $(\mathbf{0}, \mathbf{0}, f^*) \in \partial C$, i.e. there exists $(\boldsymbol{\mu}, \boldsymbol{\lambda}, \alpha) \neq \mathbf{0}$ s.t. for all $(\mathbf{u}, \mathbf{v}, t) \in C$,

$$(\boldsymbol{\mu}, \boldsymbol{\lambda}, \alpha) \cdot (\mathbf{u}, \mathbf{v}, t) = \boldsymbol{\mu}^T \mathbf{u} + \boldsymbol{\lambda}^T \mathbf{v} + \alpha t \geq (\boldsymbol{\mu}, \boldsymbol{\lambda}, \alpha) \cdot (\mathbf{0}, \mathbf{0}, f^*) = \alpha f^*$$

Proof (cont'd)

4. Since $\mathbf{u}, t$ can be arbitrarily large for $(\mathbf{u}, \mathbf{v}, t) \in C$, letting $\mathbf{u}, t \rightarrow \infty$ yields $\boldsymbol{\mu} \geq \mathbf{0}, \alpha \geq 0$.
5. Since $(\mathbf{g}(\mathbf{x}), \mathbf{h}(\mathbf{x}), f(\mathbf{x})) \in C$,

$$\boldsymbol{\mu}^T \mathbf{g}(\mathbf{x}) + \boldsymbol{\lambda}^T \mathbf{h}(\mathbf{x}) + \alpha f(\mathbf{x}) \geq \alpha f^*, \quad \forall \mathbf{x} \in D$$

6. If $\alpha \neq 0$, then

$$f^* \leq f(\mathbf{x}) + (\boldsymbol{\lambda}^*)^T \mathbf{h}(\mathbf{x}) + (\boldsymbol{\mu}^*)^T \mathbf{g}(\mathbf{x}) = \mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}^*, \boldsymbol{\mu}^*)$$

where $\boldsymbol{\mu}^* = \boldsymbol{\mu}/\alpha \geq \mathbf{0}$ and $\boldsymbol{\lambda}^* = \boldsymbol{\lambda}/\alpha$. Minimizing over $\mathbf{x}$,

$$f^* \leq \phi(\boldsymbol{\lambda}^*, \boldsymbol{\mu}^*) \leq \phi^*$$

Weak duality then implies

$$f^* = \phi(\boldsymbol{\lambda}^*, \boldsymbol{\mu}^*) = \phi^*$$

i.e. strong duality holds, and the dual optimal is attained.

Proof (cont'd)

7. Now we show Slater's condition implies $\alpha \neq 0$. Suppose $\alpha = 0$.

7.1 By 5,

$$\mu^T g(x) + \lambda^T h(x) \geq 0, \quad \forall x \in D$$

7.2 Let x_0 satisfy Slater's condition, i.e. $x_0 \in \text{int } D$, $h(x_0) = \mathbf{0}$, and $g(x_0) < \mathbf{0}$. Since $\mu \geq \mathbf{0}$,

$$\underbrace{\mu^T g(x_0)}_{\leq 0} + \underbrace{\lambda^T h(x_0)}_{=0} \geq 0 \implies \mu = \mathbf{0}$$

7.3 By 7.1 and 7.2,

$$\lambda^T h(x) \geq 0, \quad \forall x \in D$$

7.4 Since $x_0 \in \text{int } D$, there exists $\delta > 0$ s.t. $x_0 + z \in D$ for all $z \in B(\mathbf{0}, \delta)$. Recalling $h(x) = Ax - b$ and $h(x_0) = \mathbf{0}$, we have

$$\lambda^T A z = \lambda^T [h(x_0 + z) - h(x_0)] \geq 0, \quad \forall z \in B(\mathbf{0}, \delta) \implies A^T \lambda = \mathbf{0}$$

7.5 Since A has full row rank, $\lambda = \mathbf{0}$. Thus $(\mu, \lambda, \alpha) = \mathbf{0}$, contradicting $(\mu, \lambda, \alpha) \neq \mathbf{0}$ given by 3.

Example

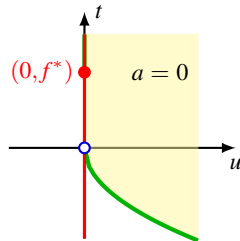
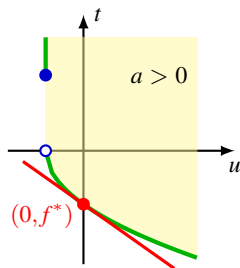
$$\min_{x \in \mathbb{R}} f(x) = \begin{cases} -\sqrt{x}, & x > 0 \\ 1 & x = 0 \\ +\infty, & x < 0 \end{cases}$$

$$\text{s.t. } x \leq a$$

is a convex problem with domain $D = [0, \infty)$.

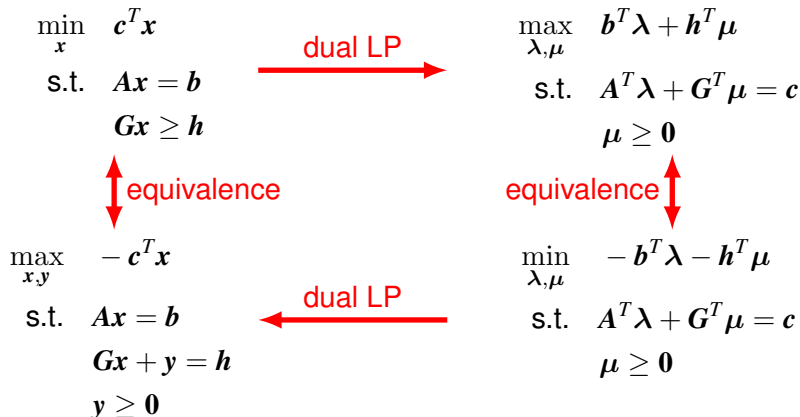
Note $\text{int } D = (0, \infty)$.

- If $a > 0$, Slater's condition is satisfied, e.g. $\frac{a}{2} \in \text{int } D$ and $\frac{a}{2} < a$, so strong duality must hold.
- If $a = 0$, no point in $\text{int } D$ is feasible. Slater's Theorem is not applicable², and it turns out that strong duality does not hold.



²Slater's condition is only a sufficient condition for strong duality. It is not necessary.

Example: Strong duality for LP



- Essentially, dual of dual is primal.
- By refined Slater's condition, strong duality holds if either the primal or the dual is feasible.
- When either f^* or ϕ^* is finite, then $f^* = \phi^*$ and they are both attained.

Example: Strong duality for LP (cont'd)

There are four possibilities

1. Primal feasible, dual feasible, $-\infty < \phi^* = f^* < +\infty$
2. Primal feasible, dual infeasible, $f^* = \phi^* = -\infty$

$$\begin{array}{ll}\min & x_1 - 2x_2 \\ \text{s.t.} & x_1 - x_2 = 1 \\ & x_1, x_2 \geq 0\end{array}\qquad \begin{array}{ll}\max & \lambda \\ \text{s.t.} & \lambda + \mu_1 = 1 \\ & -\lambda + \mu_2 = -2 \\ & \mu_1, \mu_2 \geq 0\end{array}$$

3. Primal infeasible, dual feasible, $f^* = \phi^* = +\infty$
4. Primal infeasible, dual infeasible, $f^* = +\infty, \phi^* = -\infty$

$$\begin{array}{ll}\min & x_1 - 2x_2 \\ \text{s.t.} & x_1 - x_2 \leq 1 \\ & -x_1 + x_2 \leq -2\end{array}\qquad \begin{array}{ll}\max & -\mu_1 + 2\mu_2 \\ \text{s.t.} & -\mu_1 + \mu_2 = 1 \\ & \mu_1 - \mu_2 = -2 \\ & \mu_1, \mu_2 \geq 0\end{array}$$

Note. No duality gap in Case 2 and Case 3, but $f^* - \phi^*$ is undefined.

Example: Dual formulation of SVM

Recall the primal formulation of SVM,

$$\begin{aligned} \min_{\mathbf{w}, b, \boldsymbol{\xi}} \quad & \frac{1}{2} \|\mathbf{w}\|_2^2 + C \mathbf{1}^T \boldsymbol{\xi} \\ \text{s. t.} \quad & y_i(\mathbf{x}_i^T \mathbf{w} + b) \geq 1 - \xi_i, \quad i = 1, 2, \dots, n \\ & \boldsymbol{\xi} \geq \mathbf{0} \end{aligned}$$

where $C > 0$ is a hyperparameter, and $\mathbf{1}$ is the vector of all 1's.

- convex problem with affine constraints.
- always feasible. Indeed, given any $\mathbf{w}, b$,

$$\xi_i = [1 - y_i(\mathbf{w}^T \mathbf{x}_i + b)]^+, \quad i = 1, 2, \dots, n$$

yields a feasible solution $(\mathbf{w}, b, \boldsymbol{\xi})$, where $(x)^+ = \max\{x, 0\}$.

- strong duality holds by refined Slater's condition
- can solve the dual problem instead, which turns out to be useful!

Example: Dual formulation of SVM (cont'd)

The Lagrangian is

$$\begin{aligned}\mathcal{L}(\mathbf{w}, b, \boldsymbol{\xi}, \boldsymbol{\mu}, \boldsymbol{\alpha}) &= \frac{1}{2} \|\mathbf{w}\|_2^2 + C \mathbf{1}^T \boldsymbol{\xi} + \sum_{i=1}^n \mu_i [1 - \xi_i - y_i (\mathbf{x}_i^T \mathbf{w} + b)] - \boldsymbol{\alpha}^T \boldsymbol{\xi} \\ &= \frac{1}{2} \|\mathbf{w}\|_2^2 - \left(\sum_{i=1}^n y_i \mu_i \mathbf{x}_i \right)^T \mathbf{w} - \boldsymbol{\mu}^T \mathbf{y} b + (C \mathbf{1} - \boldsymbol{\mu} - \boldsymbol{\alpha})^T \boldsymbol{\xi} + \mathbf{1}^T \boldsymbol{\mu}\end{aligned}$$

Minimizing over $\mathbf{w}, b, \boldsymbol{\xi}$ yields the dual function ($\mathbf{w} = \sum_{i=1}^n y_i \mu_i \mathbf{x}_i$),

$$\phi(\boldsymbol{\mu}, \boldsymbol{\alpha}) = \begin{cases} \mathbf{1}^T \boldsymbol{\mu} - \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \mu_i \mu_j y_i y_j \mathbf{x}_i^T \mathbf{x}_j, & \text{if } \boldsymbol{\mu}^T \mathbf{y} = 0, C \mathbf{1} - \boldsymbol{\mu} - \boldsymbol{\alpha} = \mathbf{0} \\ -\infty, & \text{otherwise} \end{cases}$$

The dual problem is

$$\begin{aligned}\max_{\boldsymbol{\mu}, \boldsymbol{\alpha}} \quad & \phi(\boldsymbol{\mu}, \boldsymbol{\alpha}) \\ \text{s. t.} \quad & \boldsymbol{\mu} \geq \mathbf{0}, \boldsymbol{\alpha} \geq \mathbf{0}\end{aligned}$$

Example: Dual formulation of SVM (cont'd)

Making the constraints explicit, we obtain the equivalent problem,

$$\begin{aligned} \max_{\boldsymbol{\mu}, \boldsymbol{\alpha}} \quad & \mathbf{1}^T \boldsymbol{\mu} - \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \mu_i \mu_j y_i y_j \mathbf{x}_i^T \mathbf{x}_j \\ \text{s. t.} \quad & \boldsymbol{\mu}^T \mathbf{y} = 0 \\ & \boldsymbol{\mu} + \boldsymbol{\alpha} = C \mathbf{1} \\ & \boldsymbol{\mu} \geq \mathbf{0}, \boldsymbol{\alpha} \geq \mathbf{0} \end{aligned}$$

Eliminating $\boldsymbol{\alpha}$, we obtain the following dual formulation of SVM,

$$\begin{aligned} \max_{\boldsymbol{\mu}} \quad & \mathbf{1}^T \boldsymbol{\mu} - \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \mu_i \mu_j y_i y_j \mathbf{x}_i^T \mathbf{x}_j \\ \text{s. t.} \quad & \boldsymbol{\mu}^T \mathbf{y} = 0 \\ & \mathbf{0} \leq \boldsymbol{\mu} \leq C \mathbf{1} \end{aligned}$$

Can be solved efficiently by an algorithm called **Sequential Minimal Optimization (SMO)**. Also amenable to further generalization using the **kernel trick** that replaces $\mathbf{x}_i^T \mathbf{x}_j$ by a **kernel (function)** $K(\mathbf{x}_i, \mathbf{x}_j)$.

Outline

- Dual function and dual problem
- Weak and strong duality
- Slater's condition
- KKT conditions revisited

KKT conditions for convex problems

Consider a differentiable convex problem and its dual,

$$\begin{array}{ll} \min_{\mathbf{x}} & f(\mathbf{x}) \\ \text{s.t.} & \mathbf{g}(\mathbf{x}) \leq \mathbf{0} \\ & \mathbf{h}(\mathbf{x}) = \mathbf{Ax} - \mathbf{b} = \mathbf{0} \end{array} \quad \left| \quad \begin{array}{ll} \max_{\boldsymbol{\lambda}, \boldsymbol{\mu}} & \phi(\boldsymbol{\lambda}, \boldsymbol{\mu}) = \inf_{\mathbf{x}} \mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\mu}) \\ \text{s.t.} & \boldsymbol{\mu} \geq \mathbf{0} \end{array}$$

Theorem. KKT conditions hold at $\mathbf{x}^*$ with Lagrange multipliers $\boldsymbol{\lambda}^*$, $\boldsymbol{\mu}^*$,

1. (primal feasibility) $\mathbf{h}(\mathbf{x}^*) = \mathbf{0}, \mathbf{g}(\mathbf{x}^*) \leq \mathbf{0}$
2. (dual feasibility) $\boldsymbol{\mu}^* \geq \mathbf{0}$
3. (stationarity) $\nabla_{\mathbf{x}} \mathcal{L}(\mathbf{x}^*, \boldsymbol{\lambda}^*, \boldsymbol{\mu}^*) = \mathbf{0}$
4. (complementary slackness) $\mu_j^* g_j(\mathbf{x}^*) = 0, j = 1, 2, \dots, m$

if and only if all the following conditions hold,

1. strong duality holds, i.e. $f^* = \phi^*$
2. $\mathbf{x}^*$ is a primal optimal solution, i.e. $f^* = f(\mathbf{x}^*)$
3. $(\boldsymbol{\lambda}^*, \boldsymbol{\mu}^*)$ is a dual optimal solution, i.e. $\phi^* = \phi(\boldsymbol{\lambda}^*, \boldsymbol{\mu}^*)$

Proof of necessity

Assume KKT holds at $\mathbf{x}^*$ with Lagrange multipliers $\boldsymbol{\lambda}^*, \boldsymbol{\mu}^*$.

- Since $\boldsymbol{\mu}^* \geq \mathbf{0}$, $\mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}^*, \boldsymbol{\mu}^*) = f(\mathbf{x}) + (\boldsymbol{\lambda}^*)^T \mathbf{h}(\mathbf{x}) + (\boldsymbol{\mu}^*)^T \mathbf{g}(\mathbf{x})$ is convex in $\mathbf{x}$.
- The stationarity condition $\nabla_{\mathbf{x}} \mathcal{L}(\mathbf{x}^*, \boldsymbol{\lambda}^*, \boldsymbol{\mu}^*) = \mathbf{0}$ implies $\mathbf{x}^*$ is a global minimum of $\mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}^*, \boldsymbol{\mu}^*)$, i.e.

$$\mathcal{L}(\mathbf{x}^*, \boldsymbol{\lambda}^*, \boldsymbol{\mu}^*) = \inf_{\mathbf{x}} \mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}^*, \boldsymbol{\mu}^*) = \phi(\boldsymbol{\lambda}^*, \boldsymbol{\mu}^*)$$

- By primal feasibility and complementary slackness,

$$\mathcal{L}(\mathbf{x}^*, \boldsymbol{\lambda}^*, \boldsymbol{\mu}^*) = f(\mathbf{x}^*) + (\boldsymbol{\lambda}^*)^T \underbrace{\mathbf{h}(\mathbf{x}^*)}_{=\mathbf{0}} + \underbrace{(\boldsymbol{\mu}^*)^T \mathbf{g}(\mathbf{x}^*)}_{=\mathbf{0}} = f(\mathbf{x}^*)$$

so

$$f(\mathbf{x}^*) = \phi(\boldsymbol{\lambda}^*, \boldsymbol{\mu}^*)$$

- By the discussion on slide 12, $\mathbf{x}^*$ is primal optimal, $(\boldsymbol{\lambda}^*, \boldsymbol{\mu}^*)$ is dual optimal and strong duality holds.

Proof of sufficiency

Assume strong duality holds, $\mathbf{x}^*$ is primal optimal, and $(\boldsymbol{\lambda}^*, \boldsymbol{\mu}^*)$ is dual optimal. We only need to show the stationarity condition and the complementary slackness condition.

$$\begin{aligned} f^* &= \phi^* && \text{(strong duality)} \\ &= \phi(\boldsymbol{\lambda}^*, \boldsymbol{\mu}^*) && \text{(dual optimality of } (\boldsymbol{\lambda}^*, \boldsymbol{\mu}^*) \text{)} \\ &= \inf_{\mathbf{x}} \mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}^*, \boldsymbol{\mu}^*) && \text{(definition of dual function)} \\ &\leq \mathcal{L}(\mathbf{x}^*, \boldsymbol{\lambda}^*, \boldsymbol{\mu}^*) && \text{(definition of infimum)} \\ &= f(\mathbf{x}^*) + \underbrace{(\boldsymbol{\lambda}^*)^T \mathbf{h}(\mathbf{x}^*)}_{=0} + \underbrace{(\boldsymbol{\mu}^*)^T \mathbf{g}(\mathbf{x}^*)}_{\geq 0} \underbrace{\phantom{(\boldsymbol{\mu}^*)^T \mathbf{g}(\mathbf{x}^*)}}_{\leq 0} \\ &\leq f(\mathbf{x}^*) && \text{(primal and dual feasibility of } \mathbf{x}^*, \boldsymbol{\mu}^* \text{)} \\ &= f^* && \text{(primal optimality of } \mathbf{x}^* \text{)} \end{aligned}$$

So both inequality holds with equality. The first implies $\mathbf{x}^*$ is a minimum of $\mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}^*, \boldsymbol{\mu}^*)$, so $\nabla_{\mathbf{x}} \mathcal{L}(\mathbf{x}^*, \boldsymbol{\lambda}^*, \boldsymbol{\mu}^*) = \mathbf{0}$. The second implies $(\boldsymbol{\mu}^*)^T \mathbf{g}(\mathbf{x}^*) = \mathbf{0}$, so $\mu_j g_j(\mathbf{x}^*) = 0$ for $j = 1, 2, \dots, m$.

KKT conditions under Slater's condition

Corollary. If $\mathbf{x}^*$ is a minimum of a differentiable convex problem that satisfies (refined) Slater's condition, then the KKT conditions hold at $\mathbf{x}^*$ with some Lagrange multipliers λ^*, μ^* .

Proof. Under (refined) Slater's condition, strong duality holds. Since

$$\phi^* = f^* = f(\mathbf{x}^*) > -\infty$$

ϕ^* is attained at some λ^*, μ^* . By the previous theorem, the KKT conditions hold at $\mathbf{x}^*$ with λ^*, μ^* being the Lagrange multipliers.